

HW01 – QA/QC Jobs · 20 Defects · Test a Physical Product

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General Information

Exercise ID:	HW01-AI
Duration:	5 hours
Deadline:	(please see the submission link on Moodle)
Form:	Individual Assignment
Submission:	Moodle (report) + GitHub repo link (artifacts)
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AI Policy:	Open – MANDATORY declaration + attached AI Audit Report
Required Bloom-AI level:	G9.1 → G9.6 depending on HW (see 'CLO Mapping')

Outcomes

- Describe the 2026+ QA/QC job-market landscape (including AI-augmented testing roles).
- Distinguish work that AI can replace / assist / cannot replace in QA/QC.
- Design test cases for a SPECIFIC physical product (not a software product).
- 1 Understand: ask an AI Tool for an ISTQB-process mindmap and find 3 mistakes.

- 3 Analyse: analyse AI-generated test cases and find ≥ 3 missed edge cases.

Description

Note: HW01 is the warm-up — it does NOT use the EShop SUT. It focuses on the QA/QC job market, recent software defects, and testing a real physical device that you own. The EShop SUT is used from HW02 onwards.

Requirement 1 – QA/QC Job Market 2026+ (40 pts)

- Find 10 QA/QC job postings PUBLISHED WITHIN 60 DAYS of the submission date.
- Mandatory: ≥ 3 positions REQUIRING AI/LLM/automation-AI skills.
- Each posting: link, dated screenshot, job description, required skills, salary.
- Write 1–2 sentences of "AI Impact Analysis" per posting.
- Anti-cheat: screenshots must show your account name in the corner.
Clarification: "account name" refers to your login name / display name on the job platform (LinkedIn, Indeed, etc.) — it does NOT need to match your StudentID.

Requirement 2 – 20 Software Defects 2022–2026 (20 pts)

- Find 20 software defects publicized between 2022 and 2026.
- Mandatory: ≥ 5 defects related to AI/LLM (hallucination, prompt injection, bias).
- Each defect: source link, description, severity, consequences, solution.
- NEW: find 1 place where the AI is biased or hallucinates when explaining the defect. **Clarification: this applies to EVERY defect — each of the 20 entries must include 1 identified instance of AI bias or hallucination (20 instances total).**

Requirement 3 – Test cases for ONE physical product (40 pts)

- Choose a SPECIFIC household device (fan / water filter / rice cooker / smart bulb...).
- Submit 1 photo of THE DEVICE + your student ID card in the SAME frame.
- Declare brand, model, year, serial number (mask the middle 4 chars).

- Design 15 test cases (Objective / Input / Steps / Expected / Actual / Verdict).
Clarification: 15 test cases total. Execute and record videos for ≥ 5 out of the 15 (not all 15 need videos). Also aim to find ≥ 5 defects from the device during execution.
- ≥ 3 test cases must be edge cases the AI Tool could NOT find.
Clarification: students must provide BOTH (a) a screenshot of the AI conversation showing the AI did not generate these edge cases, AND (b) a written explanation of why the AI missed them.
- Execute ≥ 5 test cases on the real device and record short videos (≤ 60 s).

AI Collaboration Protocol (Mandatory)

This block applies to ALL AI-first homework. Failure to comply with any item below will result in deductions per the rubric.

1. Mapping to Course Learning Outcomes (CLOs / Bloom-AI)

CLO	Short description	Required activity in this HW
G9.1	Ask AI Tool for ISTQB mindmap and correct it	R1: AI Tool draws a QA/QC role mindmap; you find 3 mistakes.
G9.3	Analyse AI output – find missing items	R3: find ≥ 3 edge cases AI missed on a real device.

2. Allowed tools & Bloom-AI level

You may use these tools (declare them in your AI Audit Report):

- Any AI Tool of your choice (e.g., ChatGPT, Claude, Gemini, Copilot, Cursor) — declare what you used.
- Faculty does NOT provide paid accounts — students choose freely.

Bloom-AI level for this homework: G9.1 (Understand) · G9.3 (Analyse).

3. AI Audit Report (attach as Appendix, mandatory)

Each AI-generated artifact (test case, script, checklist, OpenAPI spec, JMeter plan, etc.) must have an entry following the 5-section template below.

Clarification: one batch generated by a single prompt counts as ONE artifact (1 Audit Report entry) — e.g., 15 test cases produced in one prompt = 1 entry, not 15.

Item	Content
(1) Prompt + tool	Full prompt + tool name (Claude / GPT-5 / Cursor / Diffblue / Copilot...) with timestamp HH:MM dd/mm/yyyy.
(2) AI output	Full AI output (or red-bordered screenshot with annotations) – no summary, no paraphrase.
(3) Verdict	VALID / INVALID / INCOMPLETE – with reasoning grounded in ISTQB or course materials.
(4) Reasoning	2–5 sentences citing the matching slide / ISTQB section.
(5) Student fix	The corrected test case / script / checklist – highlight what changed.

At the end of the report, summarize the AI accuracy ratio (VALID / INVALID / INCOMPLETE percentages) and conclude: WHEN should AI be used / not used for this work?

4. AI Critique (200–300 words, mandatory)

Write a 200–300-word paragraph critiquing the AI. Where did the AI get it wrong, biased, or incomplete? Why did the AI fail to catch it? What principle have you learned for collaborating with AI on this HW?

5. Mandatory Disclosure

Standard template (paste at the end of your report, before appendices):

"[Test cases / script / dataset / report] was initially generated by [AI tool name]; I reviewed and modified [section X], added [edge cases Y, Z]; [section W] was written entirely by me. The detailed AI Audit Report is attached as Appendix A. I confirm I did not use AI to generate any artifact listed in the prohibited category below."

6. Anti-AI-Cheat mechanisms (CONSTRAINTS)

The artifacts below MUST NOT be AI-generated. They are evidence TAs check during grading. If any are found to be AI-generated, the entire HW receives 0 and is forwarded to the disciplinary board.

- Device photo with your student-ID card in the same frame.
- Execution videos must contain your own voice narration.
- All 10 job-posting screenshots must show your login / username.
- Prompt log .md with timestamps for every AI prompt you sent.

7. Oral Defense

A random 30% of students will be invited to a 5–7-min oral defense during the week following the deadline. Topics: (a) run any test case / scenario live on your machine, (b) explain why you chose input X and not input Y, (c) point out 1 mistake the AI made that you corrected. Failing ≥ 2 questions $\rightarrow$ HW $\times 0.5$.

Required Templates (from /AI Templates folder)

Every AI-assisted submission MUST attach the following templates from the 'AI Templates' folder: (i) [AI-02] AI Audit Report (5-section per artifact); (ii) [AI-03] AI Disclosure Form (signed); (iii) [AI-05] Privacy & Responsible Use Checklist (signed). Major Projects also need [AI-04] Reflective Statement. The [AI-06] Student Acknowledgement must be signed in Week 1 and is required before ANY AI-assisted homework is graded.

Submission regulations

- Filename: StudentID_HW01_AI_<grade>.zip
- SelfAssessedGrades: 3 digits, [000, 100]
- Required contents inside the .zip:
- Main report (PDF), must contain "AI Audit Report" + "AI Critique" + "Mandatory Disclosure" sections.
- Appendix A: full prompt log (.md or .txt) with timestamps.
- Excel: Test Cases / Checklist / Test Summary Report (incremental).
- Bug screenshots: ~~FIT Mantis (with a screenshot of the Mantis homepage showing your username = StudentID)~~. **Clarification: Mantis is NO longer used for HW01. Instead, log all defects found during physical-device testing as Issues in your own GitHub repository (include screenshot of the Issues page showing your GitHub username).**
- Device photo + student-ID card (.jpg).
- YouTube Unlisted links of ≥ 5 demo videos. **Clarification: YouTube Unlisted is the primary platform. Alternative platforms (Google Drive, OneDrive, etc.) are only accepted if the YouTube video was removed/blocked due to a copyright claim — students must first make every effort to upload to YouTube.**
- QA/QC role mindmap (PNG / Markdown).

- [AI-02] AI Audit Report (5-section per artifact) — required.
- [AI-03] AI Disclosure Form — required.
- [AI-05] AI Privacy Checklist — required.
- Self-assessment section at the end of the report.

Submit to Moodle. Deadline: see the submission link.

Assessment & Self-Assessment Template

The AI-first rubric below replaces the previous rubric. Total = 100 points. Every HW has an 'AI Compliance' column worth 15–20 points – missing the Audit Report / prompt log / critique forfeits the ENTIRE column.

No.	Criteria	Grade	Self-Assessed Grade
1	Job Market 2026+ (10 jobs × 3 pts + AI Impact)	40	
2	Software Defects 2022–2026 (20 defects)	20	
3	Physical-product test design (15 TCs + 5 videos)	25	
AI-1	[AI-02] AI Audit Report (5-section) attached	8	
AI-2	AI Critique 200–300 words + [AI-03] Disclosure attached	4	
AI-3	[AI-05] Checklist signed + anti-cheat artifacts	3	
	Total	100	

References

- ISTQB Foundation Level Syllabus (latest).
- Hardman P. (2025). A Post-AI Learning Taxonomy.
- Fuster Rabella M. (2025). OECD Education Working Paper No. 338.
- Anthropic (2025). Building reliable AI test agents – engineering blog.
- DeepEval & Promptfoo docs – LLM testing frameworks.

Other regulations

- Late submission is not permitted.
- False AI disclosure = 0 grade + disciplinary board referral.
- All prompts must use the any AI tool of your choice – they are centrally logged and you are accountable.
- Copying between students (including prompts) = 0 grade for both parties.